Advanced Algebra & Systems

ANSWER KEY

Systems of three linear equations

1 Solve the system: $x + y + z = 6$, $2x - y + z = 3$, $x + 2y - z = 5$.

From the first equation, $z = 6 - x - y$. Substitute into the second equation: $2x - y + (6 - x - y) = 3$, which simplifies to $x - 2y = -3$. Substitute into the third equation: $x + 2y - (6 - x - y) = 5$, which simplifies to $2x + 3y = 11$. Solving this reduced system: multiply $x - 2y = -3$ by 2 to get $2x - 4y = -6$, then subtract from $2x + 3y = 11$: $7y = 17$, so $y = \frac{17}{7}$. Then $x = 2y - 3 = \frac{34}{7} - \frac{21}{7} = \frac{13}{7}$, and $z = 6 - x - y = \frac{42}{7} - \frac{13}{7} - \frac{17}{7} = \frac{12}{7}$.

Solving polynomial equations of higher degree

2 Solve $x^3 - 6x^2 + 11x - 6 = 0$, given that $x = 1$ is a root.

Since $x = 1$ is a root, $(x - 1)$ is a factor. Dividing: $x^3 - 6x^2 + 11x - 6 \div (x - 1) = x^2 - 5x + 6$. Factor: $(x - 2)(x - 3)$. All roots: $x = 1, 2, 3$.

Rational root theorem

3 List the possible rational roots of $2x^3 - 3x^2 - 8x + 12 = 0$, using the rational root theorem, then find one actual root by testing.

Possible rational roots are $\pm\frac{p}{q}$ where p divides 12 and q divides 2: $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12, \pm\frac{1}{2}, \pm\frac{3}{2}$. Testing $x = 2$: $2(8) - 3(4) - 8(2) + 12 = 16 - 12 - 16 + 12 = 0$ ✓. So $x = 2$ is a root.

Partial fractions

4 Express $\frac{5x - 1}{(x - 1)(x + 2)}$ in partial fractions.

$\frac{5x - 1}{(x - 1)(x + 2)} = \frac{A}{x - 1} + \frac{B}{x + 2}$. Then $5x - 1 = A(x + 2) + B(x - 1)$. At $x = 1$: $4 = 3A$, so $A = \frac{4}{3}$. At $x = -2$: $-11 = -3B$, so $B = \frac{11}{3}$. Result: $\frac{4/3}{x - 1} + \frac{11/3}{x + 2}$.

Radical equations with extraneous solutions

5 Solve $\sqrt{2x + 7} = x + 2$, and check for extraneous solutions.

Square both sides: $2x + 7 = (x + 2)^2 = x^2 + 4x + 4$, so $x^2 + 2x - 3 = 0$. Factor: $(x + 3)(x - 1) = 0$, giving $x = -3$ or $x = 1$. Check $x = -3$: $\sqrt{2(-3) + 7} = \sqrt{1} = 1$, but $x + 2 = -1$ — mismatch, so $x = -3$ is extraneous. Check $x = 1$: $\sqrt{9} = 3$, and $x + 2 = 3$ ✓. Only solution: $x = 1$.

Systems with a quadratic and a linear equation

6 Solve the system: $y = x^2 - 3x - 4$ and $y = 2x - 4$.

Set equal: $x^2 - 3x - 4 = 2x - 4$, so $x^2 - 5x = 0$, giving $x(x - 5) = 0$, so $x = 0$ or $x = 5$. Corresponding y-values: for $x = 0$, $y = -4$; for $x = 5$, $y = 6$. Solutions: $(0, -4)$ and $(5, 6)$.

Logarithmic and exponential equations

7 Solve $\log_2(x + 3) = 5$.

Convert to exponential form: $x + 3 = 2^5 = 32$, so $x = 29$.

8 Solve $3^{2x-1} = 27$.

Since $27 = 3^3$: $2x - 1 = 3$, so $x = 2$.

Inequalities involving polynomials

9 Solve $x^2 - x - 6 > 0$, giving your answer using interval notation reasoning.

Factor: $(x - 3)(x + 2) > 0$. This is positive when both factors are positive ($x > 3$) or both are negative ($x < -2$). Solution: $x < -2$ or $x > 3$.

A multi-part system and polynomial combination problem

10 This question has two parts. A cubic polynomial $f(x) = x^3 + ax^2 + bx - 6$ has $x = 1$ and $x = 2$ as roots. (a) Use the factor theorem to set up and solve a system for a and b. (b) Find the third root of the polynomial.

(a) $f(1) = 0$: $1 + a + b - 6 = 0$, so $a + b = 5$. $f(2) = 0$: $8 + 4a + 2b - 6 = 0$, so $4a + 2b = -2$, i.e. $2a + b = -1$. Subtracting $a + b = 5$ from $2a + b = -1$: $a = -6$, then $b = 5 - (-6) = 11$. (b) With $a = -6, b = 11$: $f(x) = x^3 - 6x^2 + 11x - 6$. Since $x = 1$ and $x = 2$ are roots, dividing out $(x - 1)(x - 2) = x^2 - 3x + 2$ from the cubic gives a remaining linear factor $(x - 3)$, so the third root is $x = 3$.

Synthetic division

11 Use synthetic division to divide $2x^3 - 5x^2 + 3x - 1$ by $(x - 2)$, and state the quotient and remainder.

Using synthetic division with root 2: coefficients $2, -5, 3, -1$. Bring down 2. $2 \times 2 = 4$; $-5 + 4 = -1$. $-1 \times 2 = -2$; $3 + (-2) = 1$. $1 \times 2 = 2$; $-1 + 2 = 1$. Quotient: $2x^2 - x + 1$, remainder: 1.

Nonlinear systems with three variables

12 Solve the system: $x^2 + y^2 = 25$ and $y = x + 1$.

Substitute: $x^2 + (x+1)^2 = 25$, so $x^2 + x^2 + 2x + 1 = 25$, giving $2x^2 + 2x - 24 = 0$, or $x^2 + x - 12 = 0$. Factor: $(x+4)(x-3) = 0$, so $x = -4$ or $x = 3$. Corresponding y-values: for $x = -4$, $y = -3$; for $x = 3$, $y = 4$. Solutions: $(-4, -3)$ and $(3, 4)$.

Simplifying complex rational expressions

13 Simplify $\dfrac{\frac{1}{x} + \frac{1}{y}}{\frac{1}{x} - \frac{1}{y}}$.

Multiply numerator and denominator by xy: $\dfrac{y + x}{y - x}$.

14 Simplify $\dfrac{x^2 - 4}{x^2 - 5x + 6}$.

Factor both: $\dfrac{(x-2)(x+2)}{(x-2)(x-3)} = \dfrac{x+2}{x-3}$ (for $x \neq 2$).

Solving equations with rational exponents

15 Solve $x^{2/3} = 9$ for x.

Raise both sides to the power $\frac{3}{2}$: $x = 9^{3/2} = (9^{1/2})^3 = 3^3 = 27$.

A multi-part rational function and system problem

16 This question has three parts. A boat's fuel efficiency is modeled by $E(v) = \dfrac{600}{v} + 0.02v^2$, where v is speed in mph and E is fuel used (gallons) for a fixed trip. (a) Find $E(20)$. (b) Find $E(30)$. (c) Based on parts (a) and (b), determine whether increasing speed from 20 to 30 mph increases or decreases fuel usage for this trip, and explain the two competing effects in the formula that cause this.

(a) $E(20) = \dfrac{600}{20} + 0.02(400) = 30 + 8 = 38$ gallons. (b) $E(30) = \dfrac{600}{30} + 0.02(900) = 20 + 18 = 38$ gallons. (c) Interestingly, fuel usage is the same (38 gallons) at both speeds — increasing speed decreases the time-based term $\dfrac{600}{v}$ (less time traveling means less fuel from that factor) but increases the resistance-based term $0.02v^2$ (higher speed means more fuel burned fighting drag); in this case the two effects happen to exactly balance between 20 and 30 mph.

Solving a system involving a rational equation

17 Solve the system: $\dfrac{1}{x} + \dfrac{1}{y} = \dfrac{5}{6}$ and $x + y = 5$.

From the second equation, $y = 5 - x$. Substitute into the first: $\dfrac{1}{x} + \dfrac{1}{5-x} = \dfrac{5}{6}$. Combine the left side: $\dfrac{(5-x)+x}{x(5-x)} = \dfrac{5}{x(5-x)} = \dfrac{5}{6}$. Cross-multiply: $5 \cdot 6 = 5 \cdot x(5-x)$, so $30 = 5x(5-x)$, giving $6 = x(5-x) = 5x - x^2$, so $x^2 - 5x + 6 = 0$. Factor: $(x-2)(x-3) = 0$, giving $x = 2$ or $x = 3$. Corresponding y-values: for $x = 2$, $y = 3$; for $x = 3$, $y = 2$. Solutions: $(2, 3)$ and $(3, 2)$.